

Voice Creator – Modular Text-to-Speech Application

Project Idea Report

Name: Rajat Prakash Dhal

USN: 1HK22CS115

Email: 1hk22cs115@hkbk.edu.in

1. Introduction

This report presents the idea for a side project titled “**Voice Creator**”, a modular text-to-speech (TTS) application. The project focuses on providing users with full control over voice generation, enabling customization beyond the limitations of traditional TTS systems.

2. Problem Statement

Existing text-to-speech applications offer limited voice options with minimal customization. Users typically cannot modify important voice attributes such as pitch, tone, speed, or emotional style in a flexible way. This lack of control creates challenges for:

- Content creators who need unique narration styles
- Accessibility users who require clear and comfortable audio
- Students and educators who benefit from personalized learning tools

There is a need for a system that allows **dynamic and modular voice customization**.

3. Proposed Solution

The proposed solution is **Voice Creator**, a mobile application that enables users to convert text into speech while customizing multiple voice parameters. The system will use a modular approach where each voice attribute can be independently adjusted, allowing users to create personalized voice outputs.

4. Target Users

The application is designed for:

- **Content creators** (video, podcast, narration)

- **Students and educators** (audio learning support)
 - **Accessibility users** (visually impaired or reading difficulties)
 - **Developers and designers** (prototype voice generation)
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5. Key Features

- **Text-to-Speech Conversion:** Convert written text into audio
 - **Voice Customization:** Adjust pitch, speed, tone, and style
 - **Real-Time Preview:** Listen instantly to changes
 - **Voice Profiles:** Save and reuse custom voice settings
 - **Audio Export:** Download and share generated audio
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6. Tools and Technologies

AI Studio

- Used for experimenting with AI-generated voices
- Helps create natural and expressive speech models
- Supports testing of different voice styles and tones

Android Studio

- Used for building the mobile application
 - Implements UI using modern frameworks (e.g., Jetpack Compose)
 - Integrates TTS APIs and manages app functionality
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7. Conclusion

Voice Creator aims to transform traditional text-to-speech systems into a **customizable and user-driven platform**. By combining AI-powered voice generation with a user-friendly mobile interface, the application provides flexibility, accessibility, and creative control for a wide range of users.